Day 5 – Session 1
ADC Design
14-Bit, 100 MSPS, Low Noise, Low Power SAR ADC

Session Speaker:

Dr. Cyril Prasanna Raj P.

Professor – Cambridge Institute of Technology

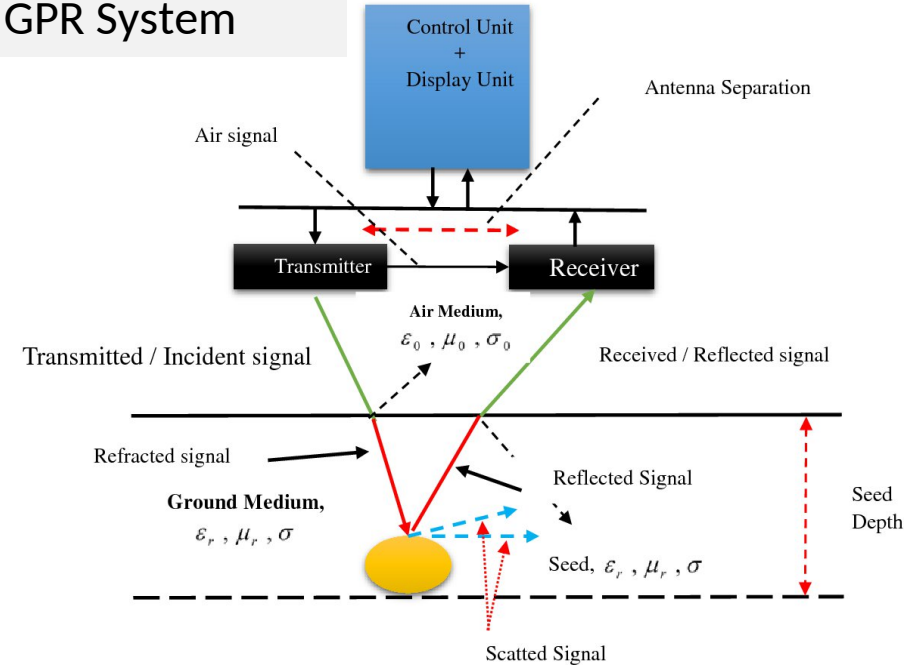
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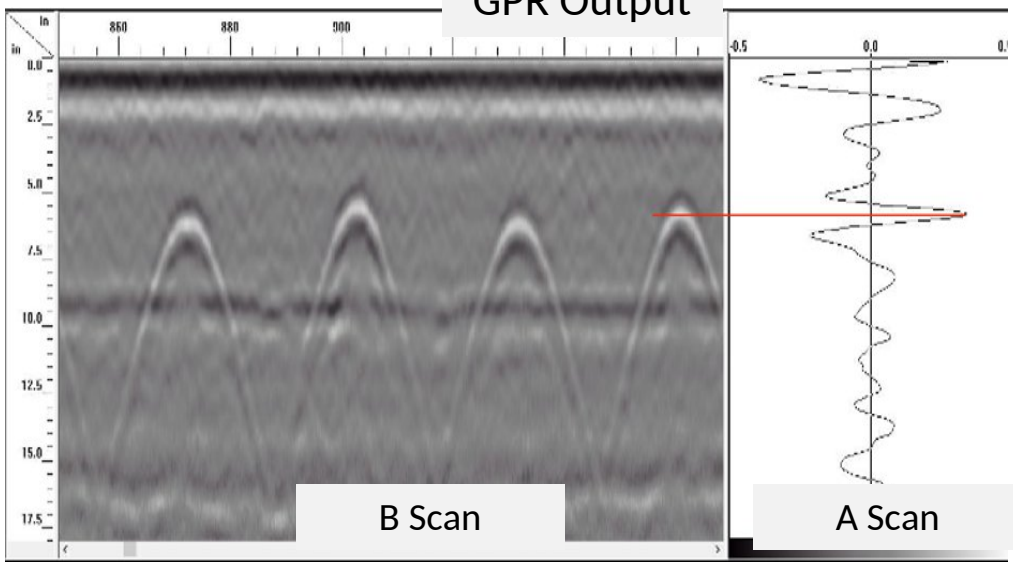
GPR Technology

GPR System



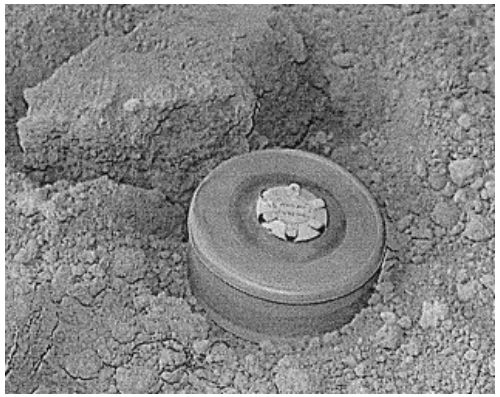


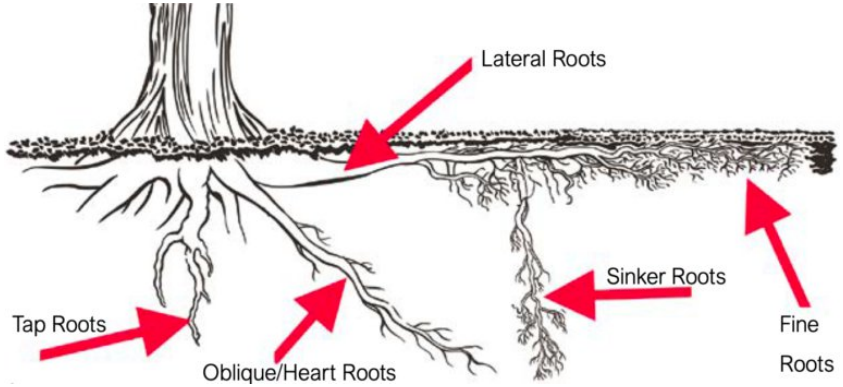
GPR Output

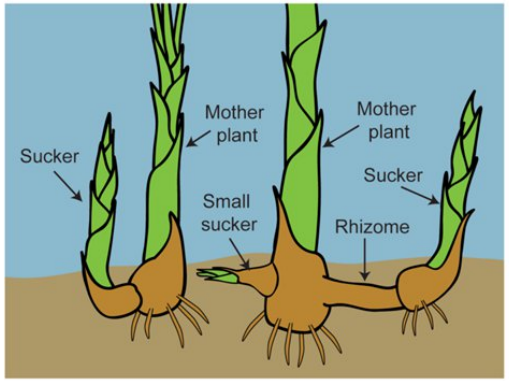


Underground Objects









Real time data acquisition





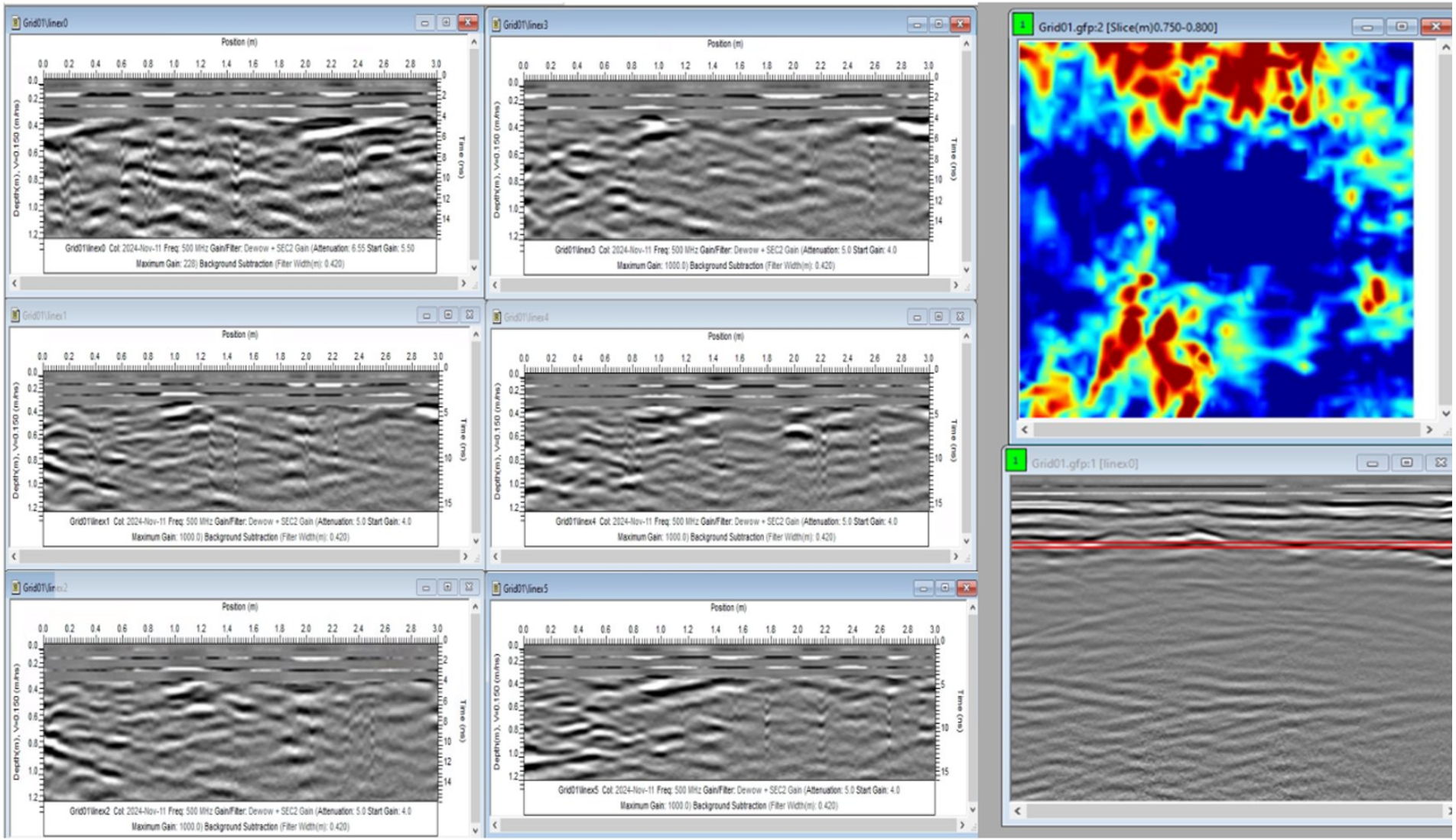




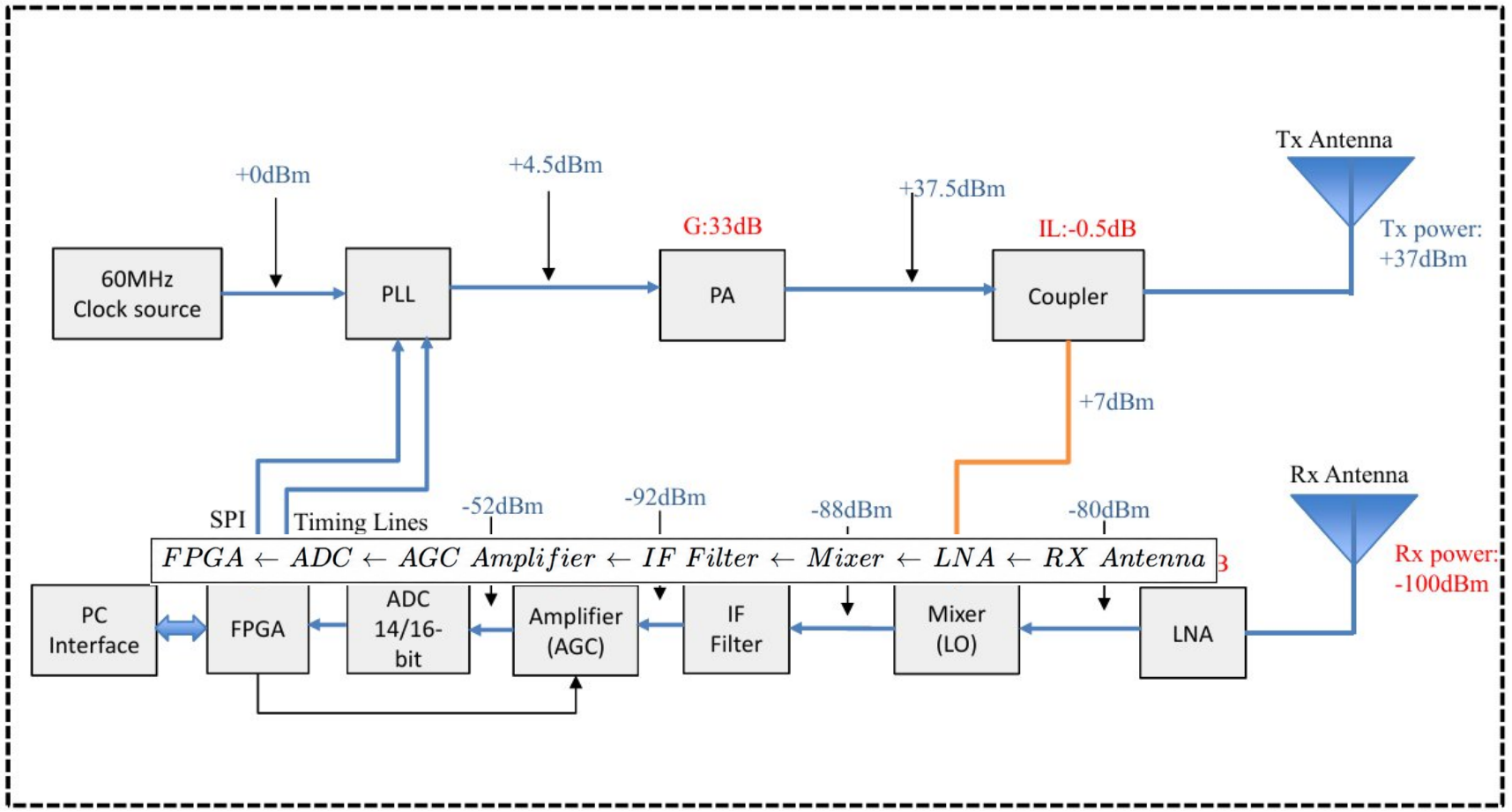




GPR Data


Grid01\linex0
Grid01\linex3
Grid01\linex1
Grid01\linex4
Grid01\linex2
Grid01\linex5
Grid01.gfp:2 [Slice(m)0.750-0.800]
Grid01.gfp:1 [linex0]

Design of ADC



Link Budget Analysis

Point	Level
RX antenna	−100 dBm
LNA output, +20 dB	−80 dBm
Mixer output, 8 dB loss	−88 dBm
IF filter output, 4 dB loss	−92 dBm
AGC amplifier, +40 dB	−52 dBm
ADC input	−52 dBm

AGC gain	ADC input
40 dB	−52 dBm
50 dB	−42 dBm
60 dB	−32 dBm
70 dB	−22 dBm
80 dB	−12 dBm

ADC Design

$$P_{ADC,in} = -52\ dBm$$

$$P_{\mathrm{dBm}} = 10\log_{10}\left(\frac{P_{\mathrm{W}}}{1\ \mathrm{mW}}\right)$$

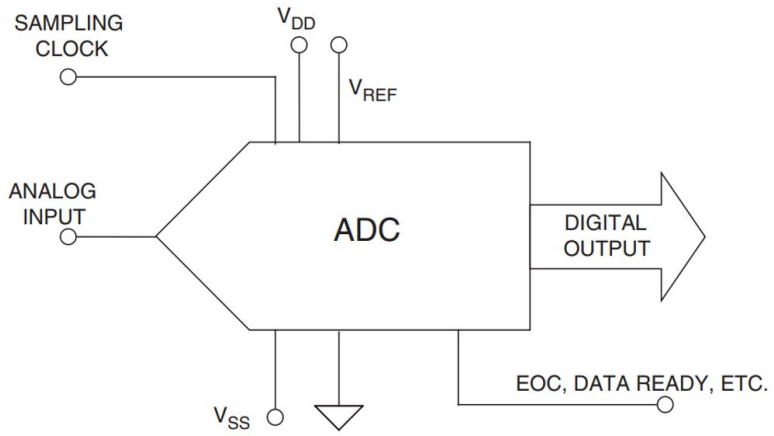
$$10^{P_{\mathrm{dBm}}/10} = \frac{P_{\mathrm{W}}}{10^{-3}}$$

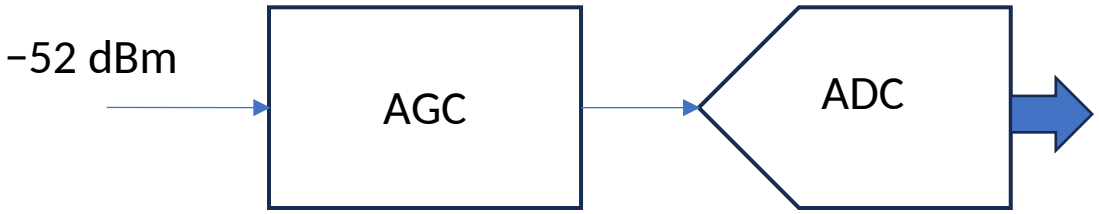
$$P = 6.31 \times 10^{-9} W$$

$$V_{rms} = \sqrt{PR}$$

$$V_{pp} = 2\sqrt{2}V_{rms}$$

$$V_{pp} \approx 1.59\ mV$$





−52 dBm signal is only about 1.59 mVpp
too small to directly drive a 1.8 V ADC efficiently

$$V_{FS} = 1.0V_{PP}$$

$$G = \frac{1.0}{1.59mV}$$

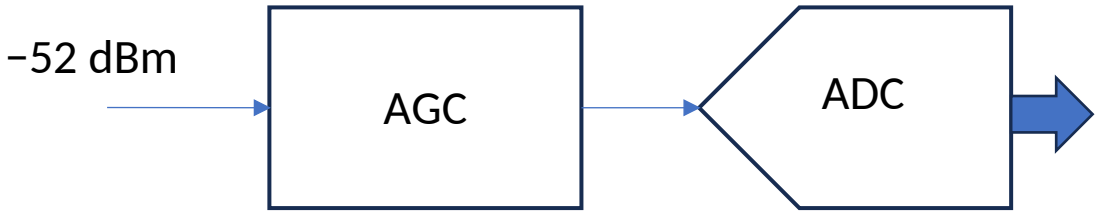
$$G = 629$$

$$G_{dB} = 20\log_{10}(629) \approx 56dB$$

AGC gain	ADC input for −52 dBm
30 dB	50.2 mVpp
40 dB	159 mVpp
50 dB	502 mVpp
56 dB	1.00 Vpp
60 dB	1.59 Vpp

ADC Design

ADC resolution	56 dB	1.00 Vpp



Assume $SNR_{quantization} = 6.02N + 1.76$

12 14 16

$SNR = 74dB$ $SNR = 86dB$ $SNR = 98dB$

$$LSB = \frac{V_{FS}}{2^N}$$

$LSB = 244.14\ \mu V$

$LSB = 61\ \mu V$

$LSB = 15.3\ \mu V$

$f_{IN} = 10MHz$

At 50 MSPS

$5\ samples/cycle$

At 100 MSPS

$10\ samples/cycle$

ADC option	VLSB	Ideal SNR	Samples/cycle	Assessment
12-bit, 50 MSPS	244.1 µV	74 dB	5	Lower complexity
12-bit, 100 MSPS	244.1 µV	74 dB	10	Good speed, limited resolution
14-bit, 50 MSPS	61.0 µV	86 dB	5	Good resolution, lower speed
14-bit, 100 MSPS	**61.0 µV**	**86 dB**	**10**	**Recommended**
16-bit, 50 MSPS	15.3 µV	98 dB	5	High resolution, harder
16-bit, 100 MSPS	15.3 µV	98 dB	10	Very demanding

ADC Design

14-bit, 100 MSPS	**61.0 µV**	**86 dB**	**10**

SFDR — Spurious-Free Dynamic Range

$$SFDR = 10\log_{10}\left(\frac{P_{signal}}{P_{largest\ spur}}\right)$$

SINAD — Signal-to-Noise-and-Distortion Ratio

SNDR - Signal-to-Noise-and-Distortion Ratio

$$\text{SNDR} \approx \text{SINAD}$$

$$SNDR = \frac{P_{signal}}{P_{noise} + P_{distortion}}$$

$$SNR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$

$$\frac{P_{signal}}{P_{noise}} = 10^{86/10} \approx 3.98 \times 10^{8}$$

$$\frac{V_{signal,rms}}{V_{noise,rms}} = 10^{86/20} \approx 19953$$

RMS signal is approximately 20,000 times larger than the RMS quantization noise for an ideal full-scale sine wave

ADC Design

$$86.04dB = \text{actual ADC SNR}$$

Noise	Source
Quantization noise	ADC conversion
Sampling kT/Cnoise	Sample-and-hold / CDAC
Comparator noise	Comparator
DAC thermal noise	Capacitive DAC
Reference noise	V_{REF}
Clock jitter noise	Sampling clock
Switch noise	Sampling network
Amplifier/driver noise	ADC input
Power-supply noise	ADC supply
Substrate noise	IC substrate
Flicker (1/f) noise	MOS devices

$$SNR_{real} < 86.04dB \qquad SNR_{real} = 75dB$$

$$ENOB = \frac{SNDR - 1.76}{6.02} \qquad ENOB \geq 12\ bits$$

$$SFDR = 20\log_{10}\left(\frac{V_{signal}}{V_{largest\ spur}}\right)$$

$$SFDR \geq 80dBc$$

ADC Design

$$THD = 20\log_{10}\left(\frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots}}{V_1}\right)$$

10-MHz fundamental

$$2f_{IN} = 20\,MHz$$

$$3f_{IN} = 30\,MHz$$

$$4f_{IN} = 40\,MHz$$

$$SNDR = 75dB$$

Assuming THD</= SNDR

$$THD \leq -75dB$$

$$V_{IN} = 1V_{PP} \qquad V_{1,rms} = \frac{1}{2\sqrt{2}} = 353.6mV$$

$$THD = -75dB$$

Total RMS harmonic voltage

$$V_H = V_1 \times 10^{-75/20}$$

$$V_H \approx 62.9\mu V_{rms}$$

Combined RMS voltage of all harmonics should be $63\mu V_{rms}$

$$1LSB = 61.04\mu V$$

$$DNL \leq \pm 0.5LSB$$ No missing codes

$$INL \leq \pm 1LSB$$

Final ADC Specification

ADC Specification	Symbol	Target / Value
ADC Architecture	—	SAR ADC
Resolution	N	14 bits
Sampling Rate	f_s	100 MSPS
Input Signal Frequency	f_{IN}	10 MHz
Full-Scale Input	V_{FS}	1 V
LSB Voltage	V_{LSB}	61.04 µV
Ideal Quantization SNR	SNR_q	86.04 dB
Target SNR	SNR_{target}	≥ 75 dB
Target SNDR/SINAD	$SNDR$	≥ 72–75 dB
Total Harmonic Distortion	THD	≤ −75 dB
Spurious-Free Dynamic Range	SFDR	≥ 80 dBc
Effective Number of Bits	ENOB	≥ 12 bits
Differential Nonlinearity	DNL	≤ ±0.5 LSB
Integral Nonlinearity	INL	≤ ±1 LSB
Reference Voltage	V_{REF}	~1.0 V
Sampling Clock	f_{CLK}	100 MHz
Digital Output	—	14-bit CMOS
Power Consumption	P_{ADC}	< 150 mW target
Operating Temperature	T_{OP}	−40 to +85 °C
GPR Receiver Input	P_{IN}	−52 dBm

Specification and its circuit block mapping

ADC specification	ADC subsystem	Design considerations
14-bit resolution	R–2R DAC + comparator + sampling	R/2R matching, resistor noise, comparator noise, settling
100 MSPS	Sampling switch + R–2R DAC + comparator + SAR	Acquisition time, R–2R settling, comparator decision
61.04 µV LSB	R–2R DAC + reference	$V_{REF}/2^{14}$, R/2R ratio accuracy
86.04 dB ideal SNR	Quantization	Determined fundamentally by 14-bit resolution
Actual SNR ≥75 dB	R–2R ladder + sampling + comparator + reference	Total thermal and circuit noise budget
THD ≤ −75 dB	R–2R ladder + switches + reference + input driver	Resistor linearity, switch R_{ON}, reference stability
SFDR ≥80 dBc	R–2R ladder + switches + reference + comparator	R/2R matching, harmonics & spurs
DNL ≤ ±0.5 LSB	R–2R resistor matching + switches	R/2R ratio matching, switch resistance
INL ≤ ±1 LSB	R–2R resistor matching + reference + calibration	Resistor ratios, systematic mismatch, calibration
Input bandwidth ≥50 MHz	Sampling switch + input capacitance + ADC driver	R_{ON}, C_{SH}, driver bandwidth
Clock jitter <1.5 ps	Clock circuit + sampling switch	Clock source, buffer, sampling aperture
Low DAC noise	R–2R ladder + switches + reference	Resistor value, resistor thermal noise, switch noise
R–2R DAC settling	R–2R ladder + switches + parasitic capacitance	R_{EQ}, parasitic C, switch R_{ON}
Conversion accuracy	R–2R DAC + comparator	DAC settling, comparator noise/offset
Power consumption	R–2R ladder + reference + comparator + logic	R value, reference current, switching activity, tr. sizing
PVT robustness	R–2R + switches + reference + comparator	Resistor TC, R_{ON}, supply and temperature variation
Monte Carlo performance	R–2R resistor network + comparator	Resistor mismatch, device mismatch, offset

Market value for ADC

No.	Application for your SAR ADC	Global market value	India market value	Demand / growth signal
1	SFCW / Ground Penetrating Radar (GPR)	US$457.3 M (2025)	US$7.7 M (2025)	Very high — India CAGR ~11.5%
2	Radar systems / Radar IF receivers	US$96.34 B (2025)	US$1.17 B (2024)	High — India radar CAGR ~4.3%
3	Ultrasonic NDT	US$3.6 B (2025)	US$717.8 M (2025)	Very high — India NDT CAGR ~10.2%
4	Medical ultrasound	US$10.9 B (2025)	US$271.3 M (2025)	High — India CAGR ~5.4%
5	SONAR / underwater sensing	US$2.73 B (2025)	~US$160 M (2026)	High — defense/maritime demand
6	Vibration & condition monitoring	US$1.93 B (2025)	US$75.9 M (2025)	High — India CAGR ~6.1%
7	Structural Health Monitoring (SHM)	US$4.45 B (2025)	US$153.9 M (2025)	Very high — India CAGR ~21.7%
8	Industrial Data Acquisition (DAQ)	US$2.42 B (2025)	US$124.4 M (2025)	High — global CAGR ~5.7%
9	Test & Measurement	US$27.9 B (2025)	US$1.34 B (2025)	High — ~5–7% growth
10	Software Defined Radio (SDR)	US$25.44 B (2025)	US$973.8 M (2025)	High — India CAGR ~5.75%
11	Automated Test Equipment (ATE)	US$5.41–8.1 B (2025)	US$278.7 M (2025)	High — India CAGR ~11%
12	Battery Management Systems (BMS)	US$13.64 B (2025)	US$327.9–397.7 M (2025)	Very high — India ~16–35%
13	Industrial sensors	US$27.4–30.7 B (2025)	US$1.39 B (2025)	Very high — India CAGR ~9.2%
14	Smart sensors / IoT sensing	~US$70–88 B (2025)	US$3.04 B (2024)	Very high — India CAGR ~21.9%
15	Power electronics / power monitoring	US$27.23 B (2025)	US$2.85 B (2025)	High — India CAGR ~5.1%

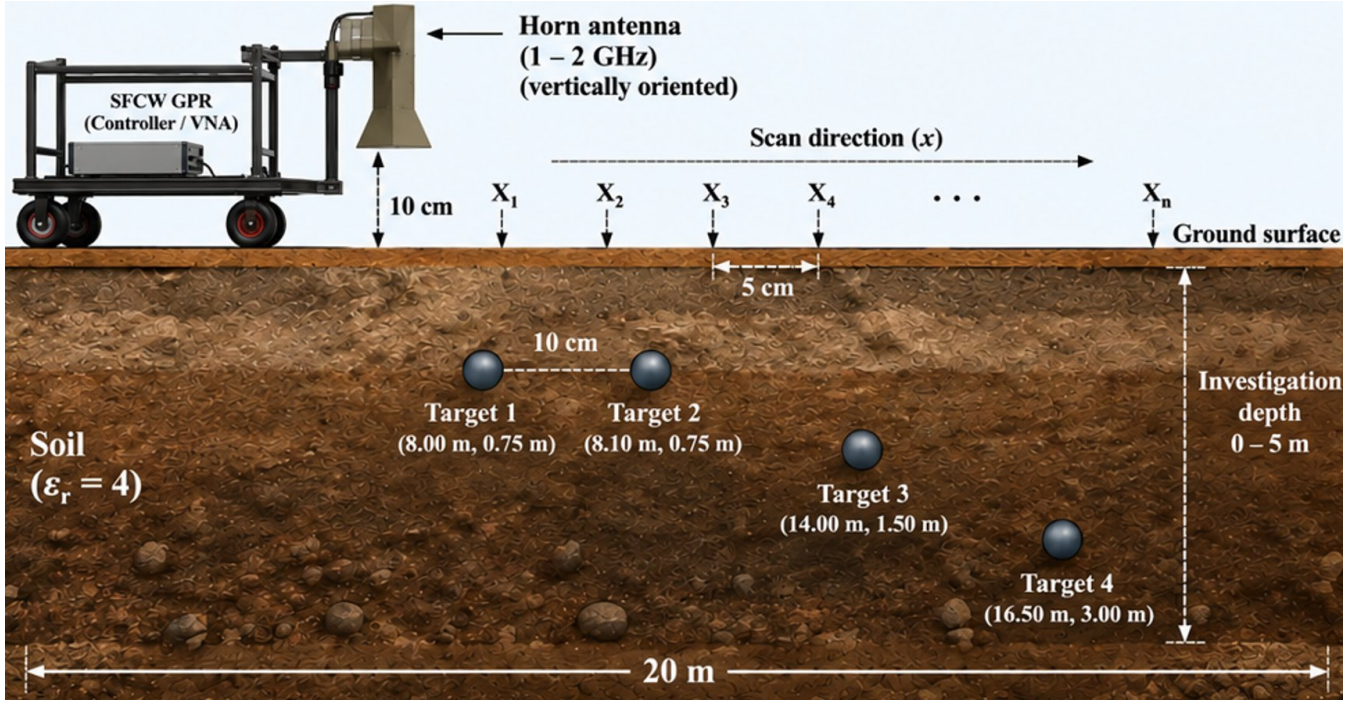
GPR Prototypes Developed

GPR Parameter	10 GHz System	1–4 GHz System
Frequency	10 GHz	1–4 GHz
Bandwidth	Design dependent	3 GHz
Centre frequency	10 GHz	2.5 GHz
Wavelength	3 cm	7.5–30 cm
Depth	Shallow	~1–3 m*
Range resolution	BW dependent	~5 cm theoretical
Scan/lateral resolution	Few cm*	~5–15 cm*
Scan sampling	≤1–2 cm*	≤5 cm*
Antenna	Parabolic dish	Horn
GPR application	High-resolution shallow sensing	Subsurface imaging





1 – 3 GHz Module with COTS Approach

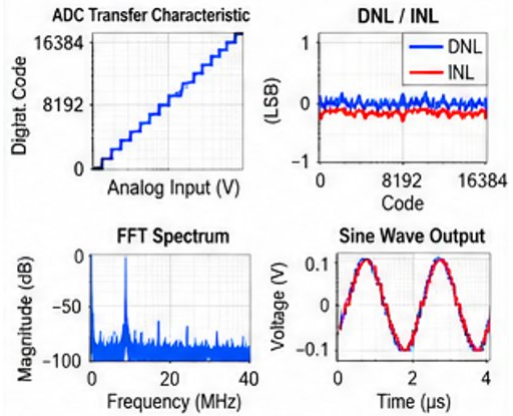


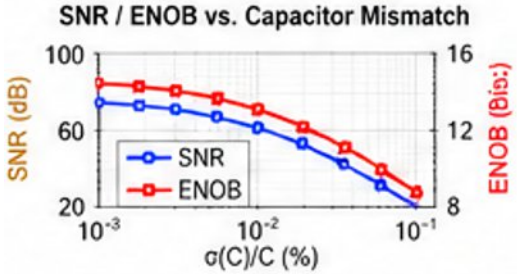


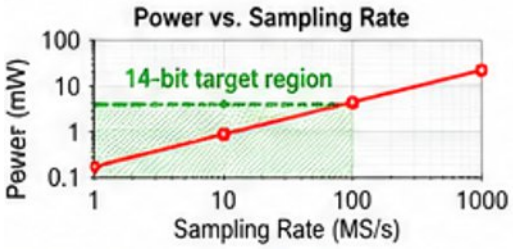


Design flow

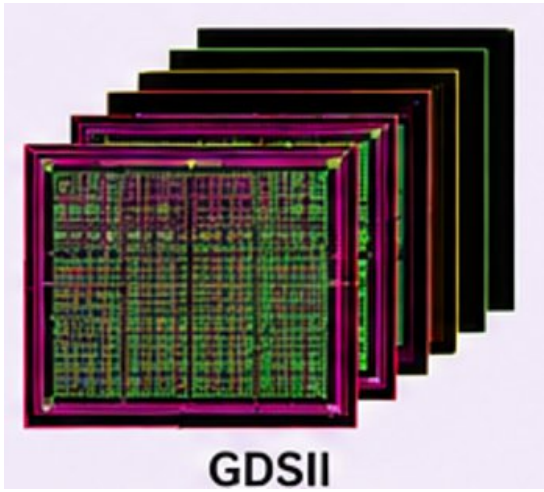
1. System-Level Specification
2. MATLAB Behavioral Modeling
3. ADC Architecture Selection
4. Behavioral Non-Ideality Modeling
5. Performance Specification
6. Circuit-Level Specification
7. Transistor-Level Design
8. Schematic Capture
9. Pre-Layout Simulation
10. PVT Corner Analysis
11. Monte Carlo Analysis
12. AMS / Mixed-Signal Verification
13. RTL SAR Logic Design
14. RTL Functional Verification
15. Analog–Digital Co-Simulation
16. Layout Floorplanning
17. Analog Layout
18. Digital Layout
19. Full ADC Top-Level Integration
20. DRC Verification
21. LVS Verification
22. Parasitic Extraction
23. Post-Layout Simulation
24. Post-Layout Performance Characterization
25. MATLAB–SPICE Correlation
26. Design Optimization / ECO
27. Final Sign-Off
28. GDSII Generation
29. GDSII Stream-Out
30. Tape-Out

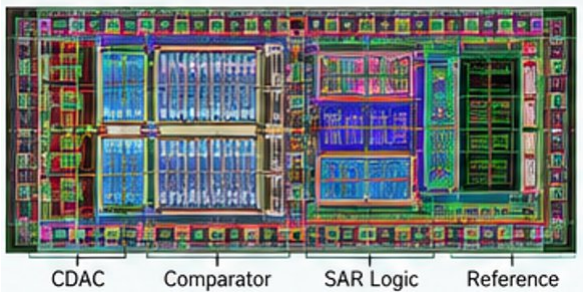






Power – Area – Speed – Accuracy – Noise – Linearity





MATLAB Modelling

Our design for ADC

$$D_{14bit} = \{D_{13:8}, D_{7:0}\}$$

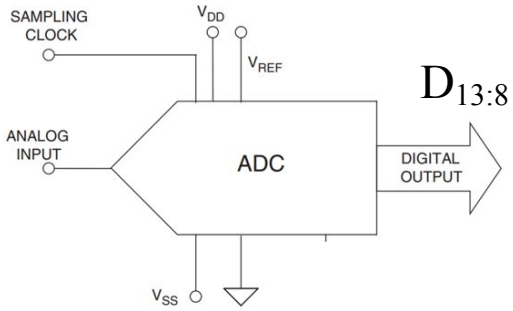
Stage 1

$$D_{13}, D_{12}, D_{11}, D_{10}, D_9, D_8$$

$$LSB_1 = 15.625\,mV$$

V_{DAC1}

6-bit R–2R DAC



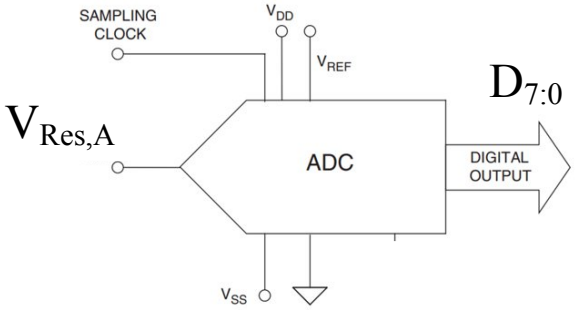
Stage 2

$$V_{RES} = V_{IN} - V_{DAC1}$$

$$V_{RES,AMP} = 256(V_{IN} - V_{DAC1})$$

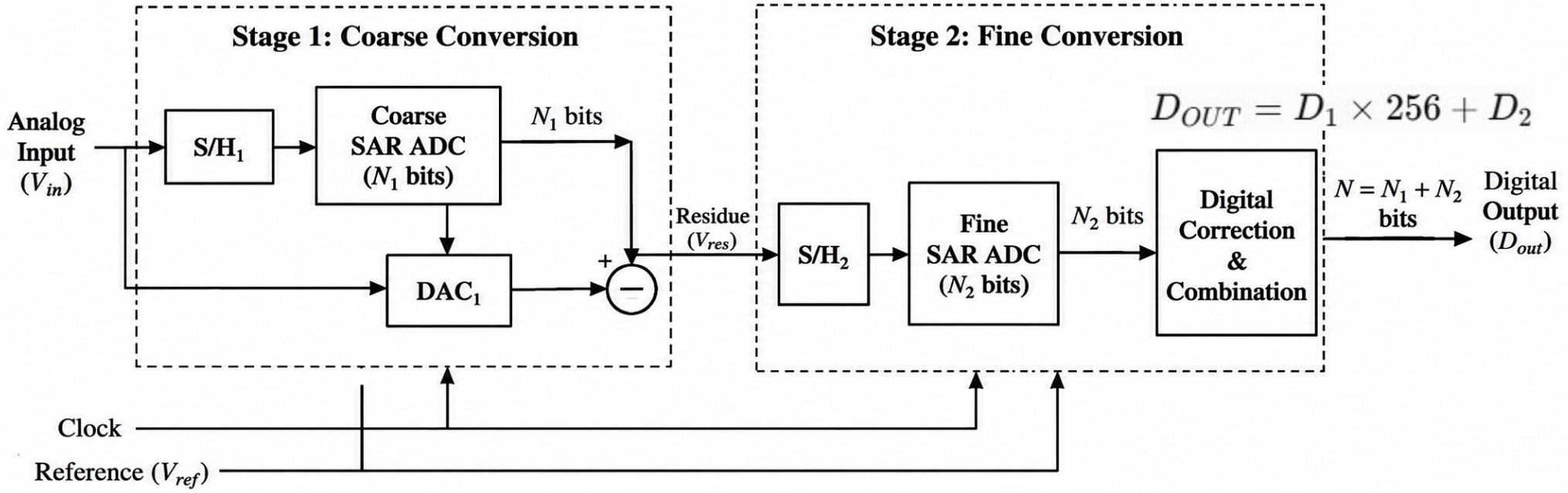
$$\frac{LSB_1}{2^8} = LSB_{14} \quad 61.04\mu V$$

8-bit R–2R DAC



Bit Allocation		
Bits	Stage	Function
D13	Stage 1	MSB
D12	Stage 1	Bit 2
D11	Stage 1	Bit 3
D10	Stage 1	Bit 4
D9	Stage 1	Bit 5
D8	Stage 1	LSB of Stage 1
D7	Stage 2	MSB of Stage 2
D6	Stage 2	Bit 2
D5	Stage 2	Bit 3
D4	Stage 2	Bit 4
D3	Stage 2	Bit 5
D2	Stage 2	Bit 6
D1	Stage 2	Bit 7
D0	Stage 2	LSB

Our design for ADC



$$V_{IN} = 0.7\,V$$

$$Code_1 = \frac{0.7}{1} \times 64$$

$$Code_1 = 44.8$$

$$44_{10} = 101100_2$$

$$D_{13:8} = 101100$$

$$V_{DAC1} = \frac{Code_1}{64} V_{REF}$$

$$= \frac{44}{64}(1) = 0.6875V$$

$$V_{IN} = 700mV$$

$$V_{DAC1} = 687.5mV$$

$$V_{RES} = V_{IN} - V_{DAC1} \qquad V_{RES} = 12.5mV$$

$$V_{RES,AMP} = 64(12.5mV) \qquad V_{IN,2} = 0.8V$$

$$Code_2 = \frac{0.8}{1} \times 256 = 204.8$$

$$204_{10} = 11001100_2$$

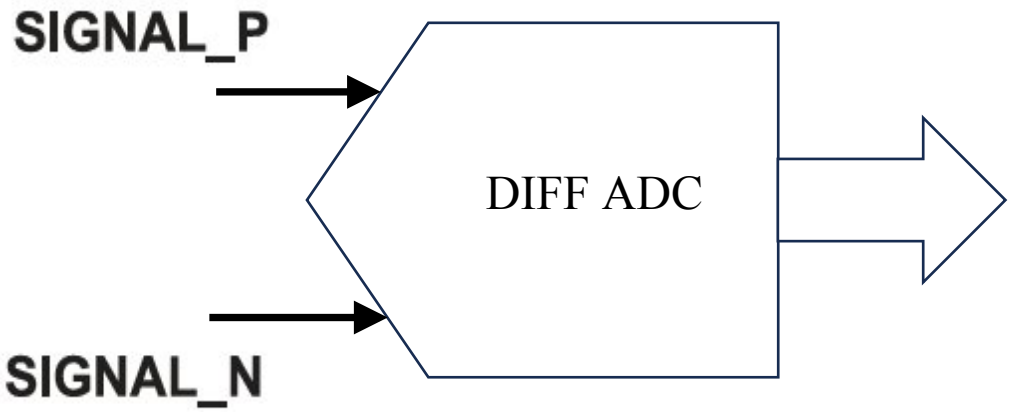
$$D_{13:0} = 10110011001100 = 11468$$

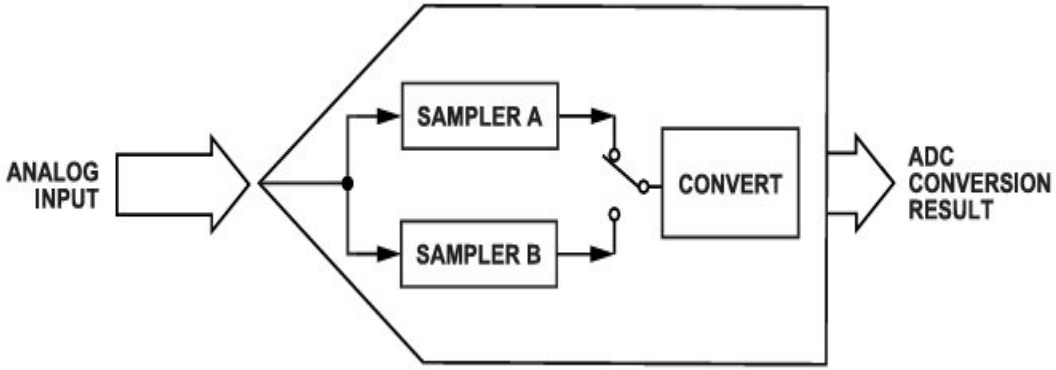
$$Code_{ideal} = 0.7 \times 16384 \qquad Code = 11468$$

Comparison

Parameter	Conventional 14-bit SAR + R–2R	Two-stage 6+8-bit SAR + R–2R	Advantage
Resolution	14 bit	14 bit	Same
Sampling rate	40 MSPS target	40 MSPS target	Same
Supply	1.8 V	1.8 V	Same
Total R–2R resistor elements	~28 minimum*	~12 + ~16 = ~28*	Similar
DAC architecture	14-bit R–2R	6-bit + 8-bit R–2R	Two-stage
Reduced DAC switching activity	Large 14-bit DAC	Smaller DACs	Potentially lower dynamic power
DAC matching requirement	Very high	Distributed	Two-stage advantage
DAC settling requirement	Very high	Lower per stage	Two-stage advantage
Reference loading	Higher instantaneous loading	Distributed	Two-stage advantage
Potential speed	Moderate	Higher potential	Two-stage
Higher conversion speed potential	One long SAR operation	Parallel/pipelined operation	Potentially higher throughput
Reconfigurability	Possible but DAC remains 14-bit	Easier to implement	Different resolution/speed/power modes
Digital correction	Minimal	Required	Single-stage simpler
Calibration requirement	Moderate	High	Single-stage advantage
Noise sources	Fewer	More	Single-stage advantage
Residue amplifier	Not required	Required	Single-stage advantage
Power	Moderate	Moderate–high	Architecture dependent
Analog complexity	Moderate	High	Single-stage advantage
Digital complexity	Low	Moderate	Single-stage advantage
Layout complexity	Moderate/high	High	Single-stage advantage
Design risk	Lower	Higher	Single-stage advantage

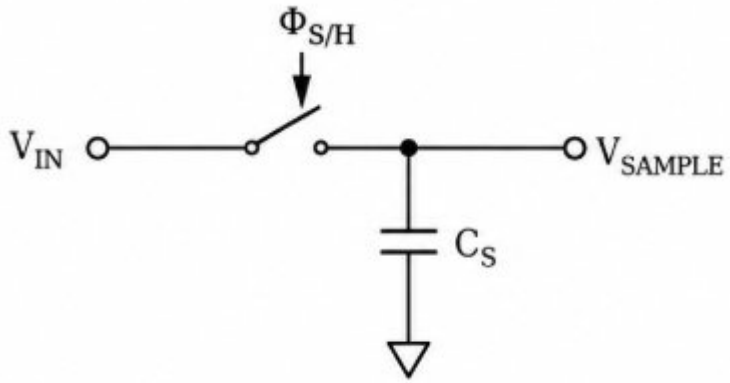
ADC Design Challenge



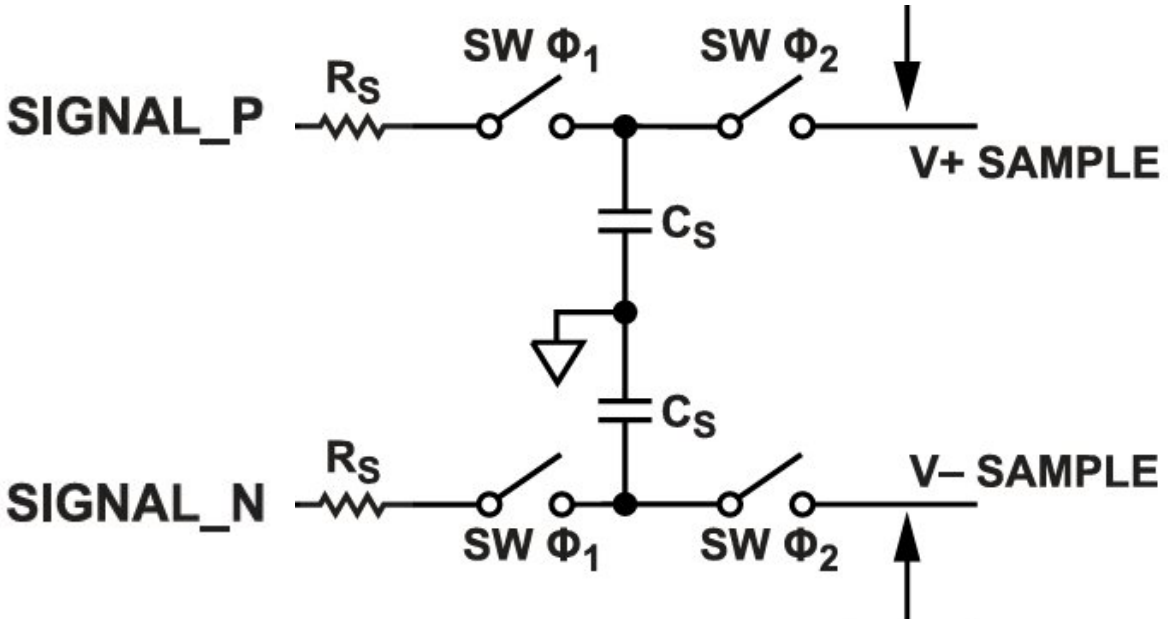


ADC Architecture
Design Challenges

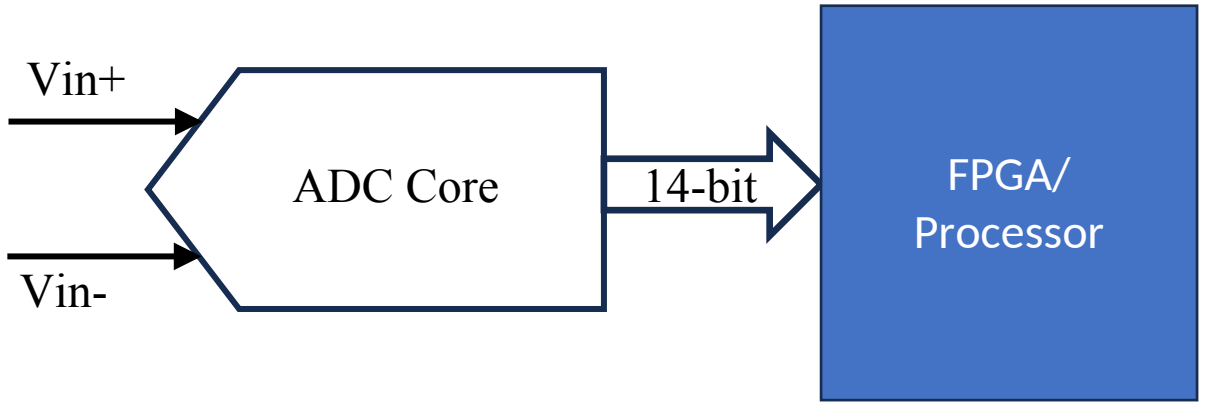
(a) Conventional Sample-and-Hold (Single-Ended)

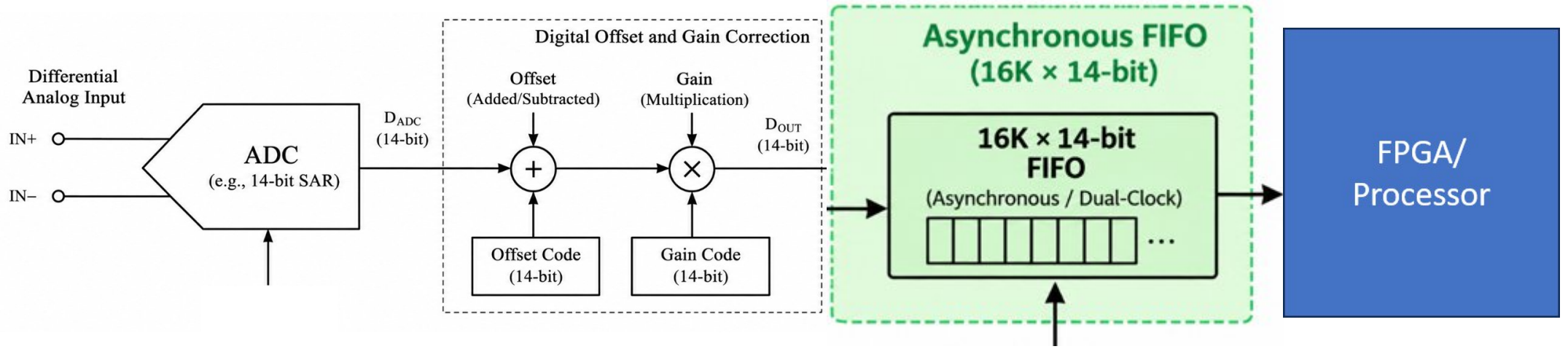


(b) Bootstrapped/Differential Sample-and-Hold

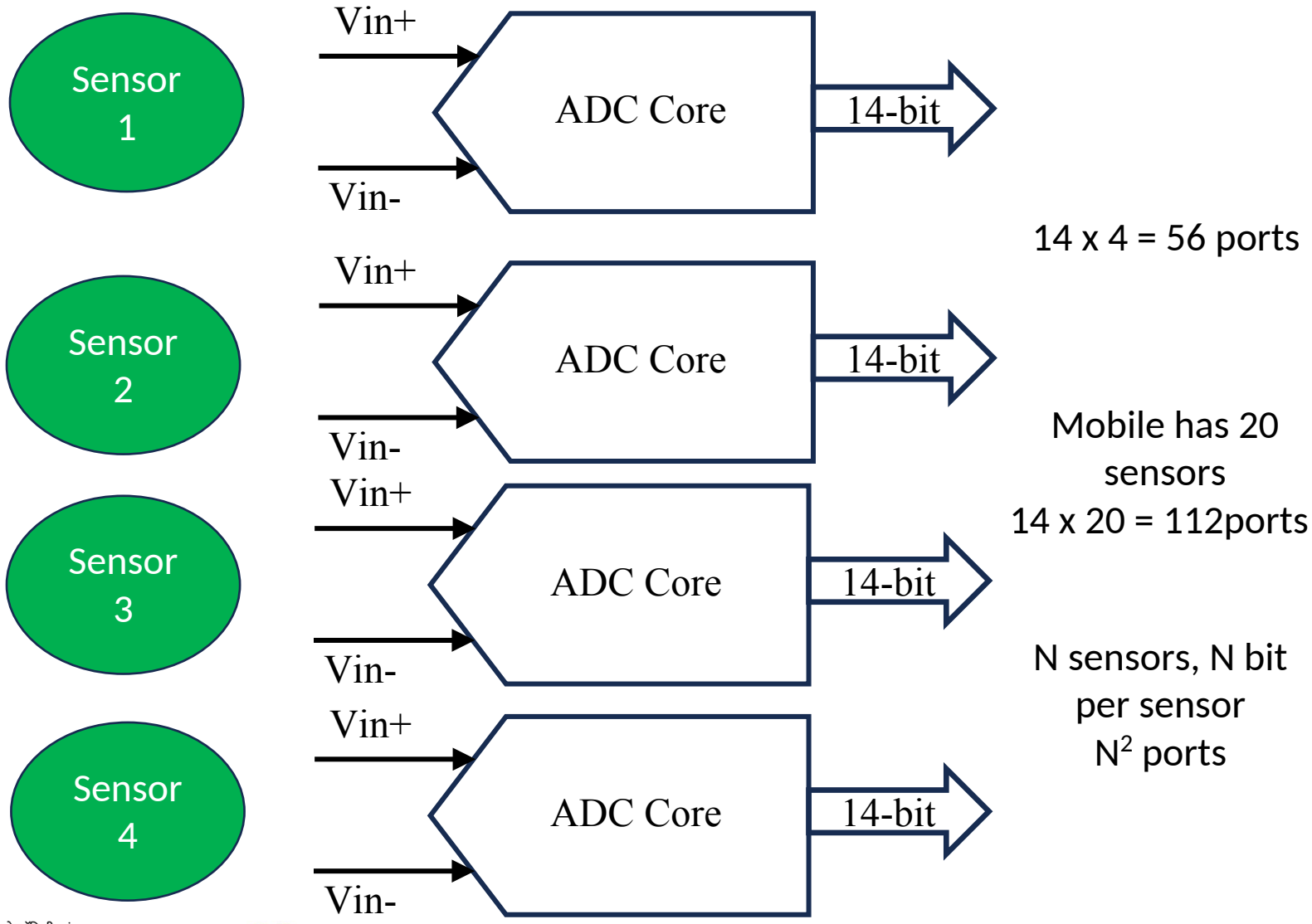


ADC Design Challenges

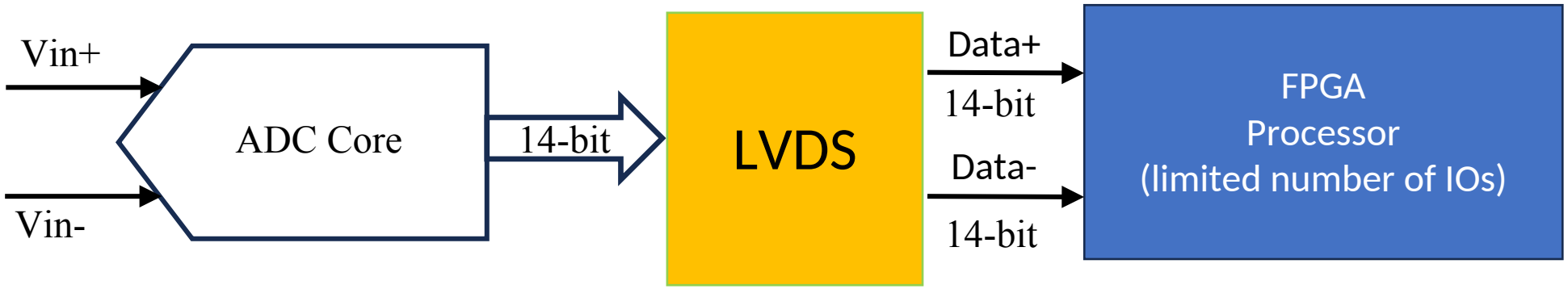




ADC Architecture Design Challenges

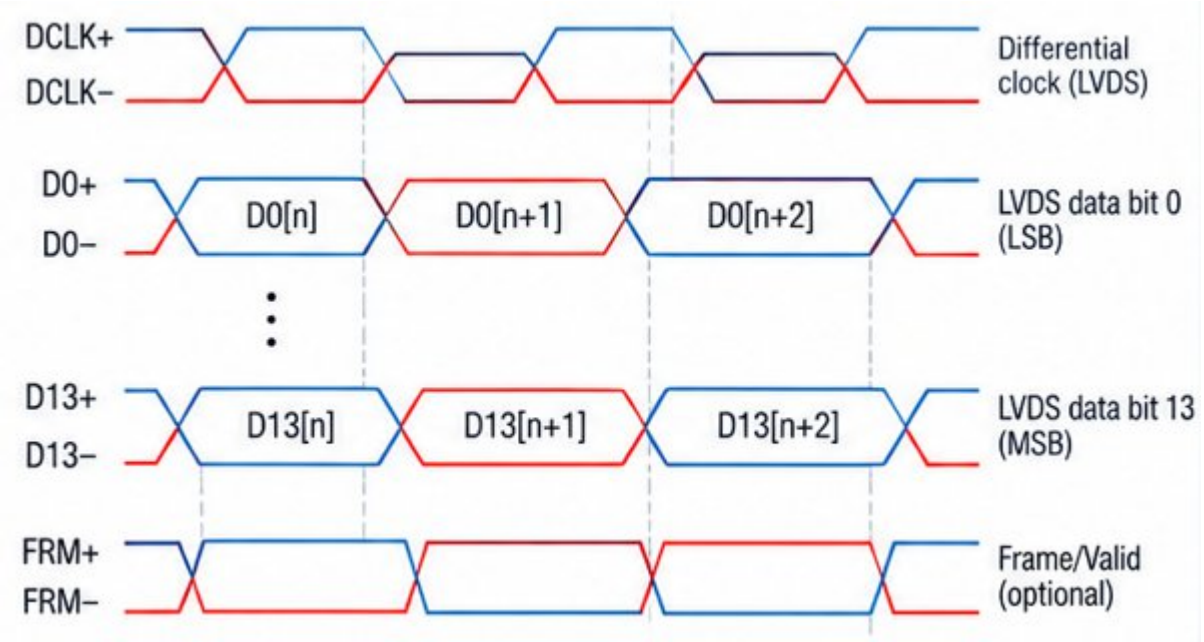


ADC Architecture Design Challenges

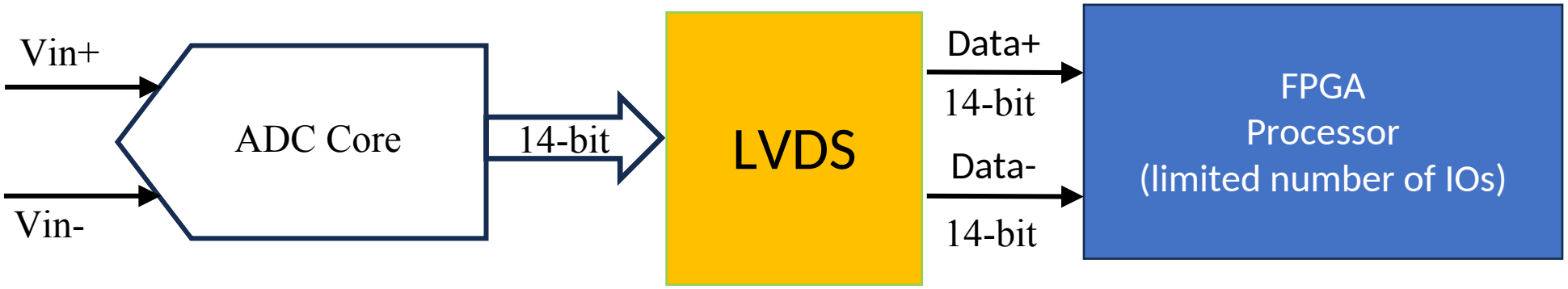


LVDS Out = V_{data+} - V_{data-} = 2Vdata

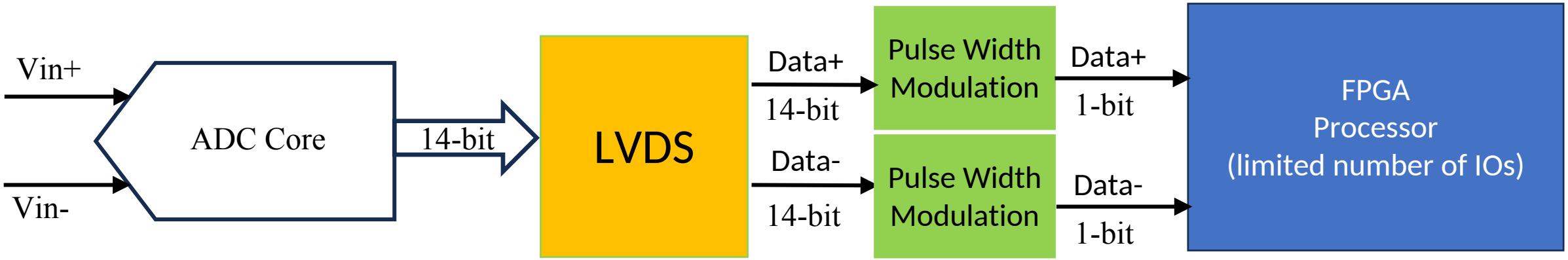
LVDS Out = V_{data+} /2 - V_{data-} /2 = Vdata



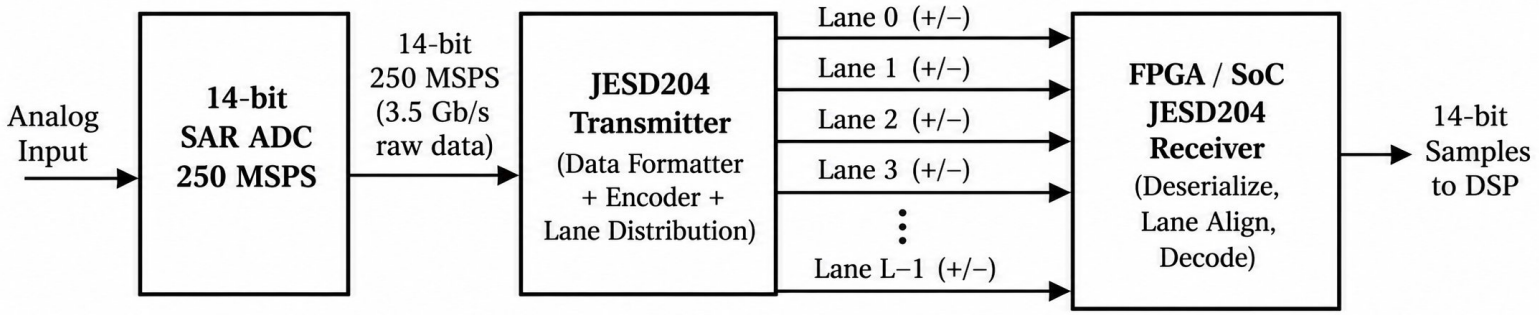
ADC Architecture Design Challenges



Any other options

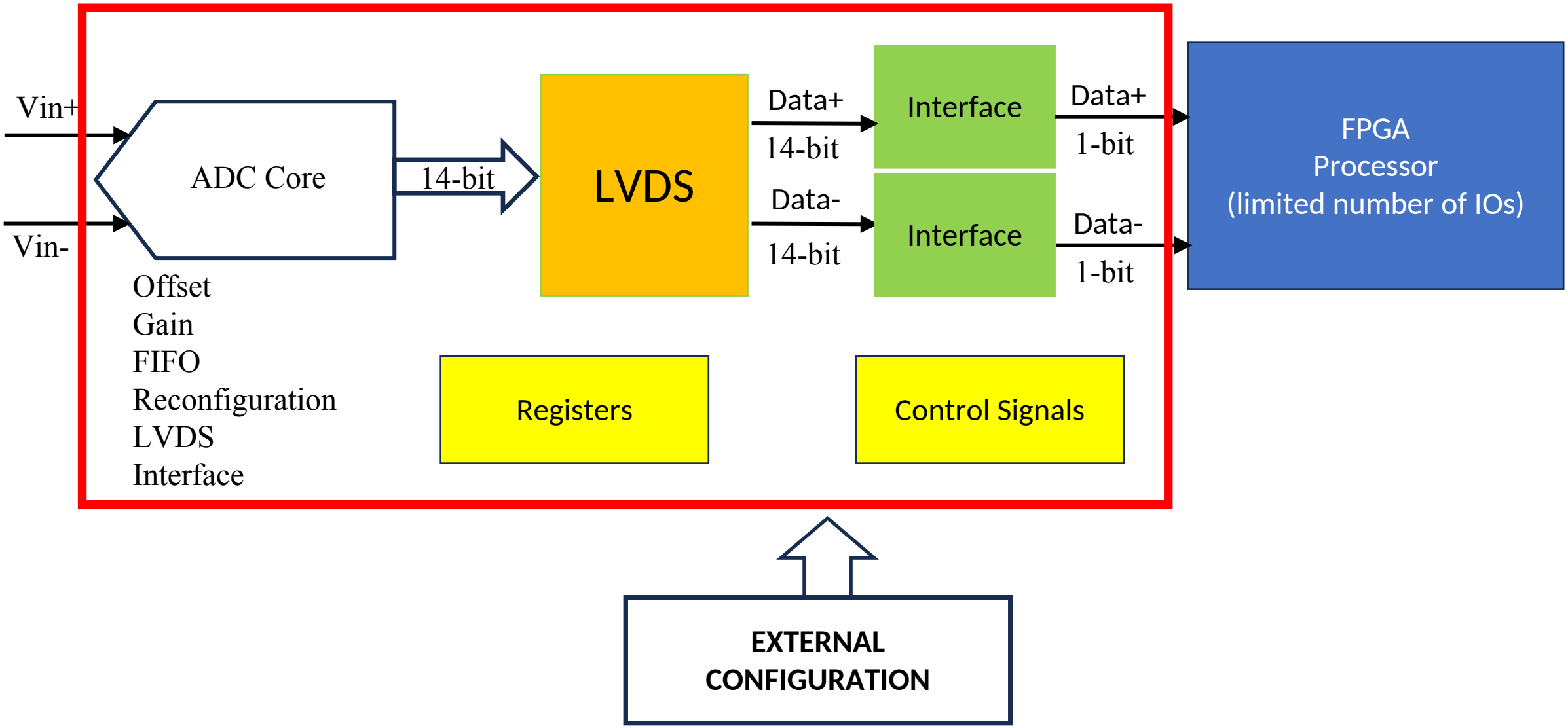


ADC Architecture Design Challenges

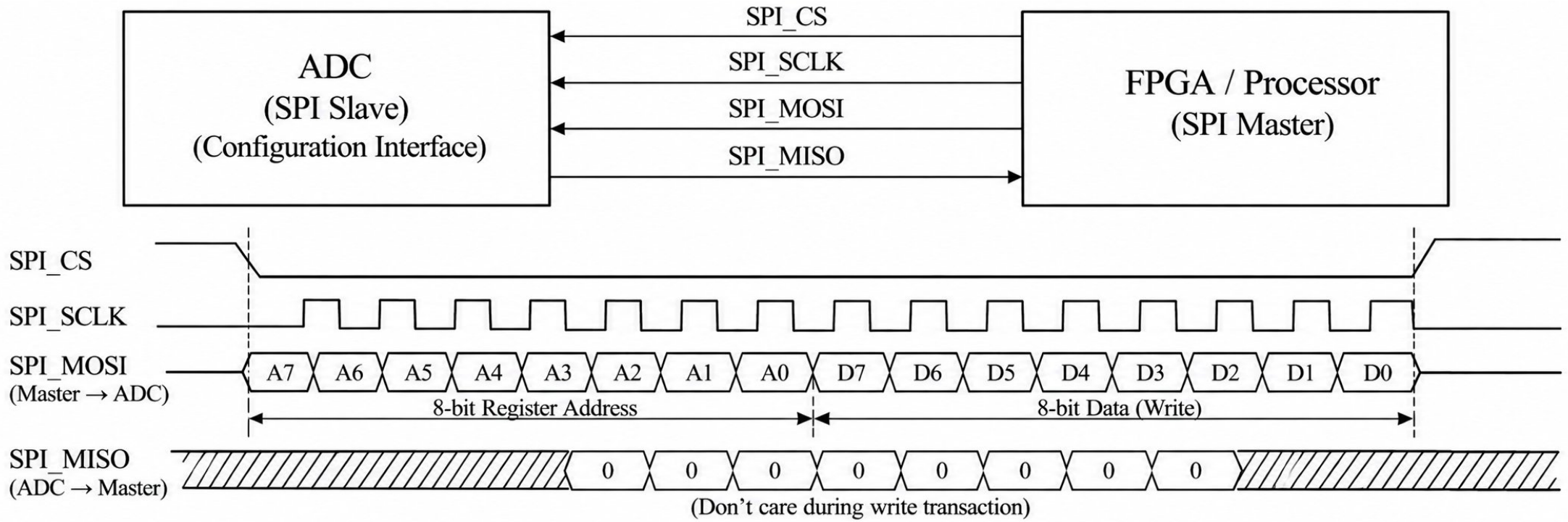


Interface	Type	Typical use
SPI	Serial, single-ended	ADC control, low/medium-speed data
I²C	Serial, single-ended	Slow sensor ADCs/configuration
Parallel CMOS	Parallel, single-ended	Low/medium-speed ADCs
LVDS	Differential	High-speed ADC data
Serial LVDS	Differential serial	High-speed ADCs
JESD204B	High-speed serial	High-performance ADC/DAC
JESD204C	High-speed serial	Modern high-speed converters
MIPI	High-speed serial	Mobile/embedded mixed-signal systems
SLVS-EC	High-speed differential	Imaging / sensor interfaces

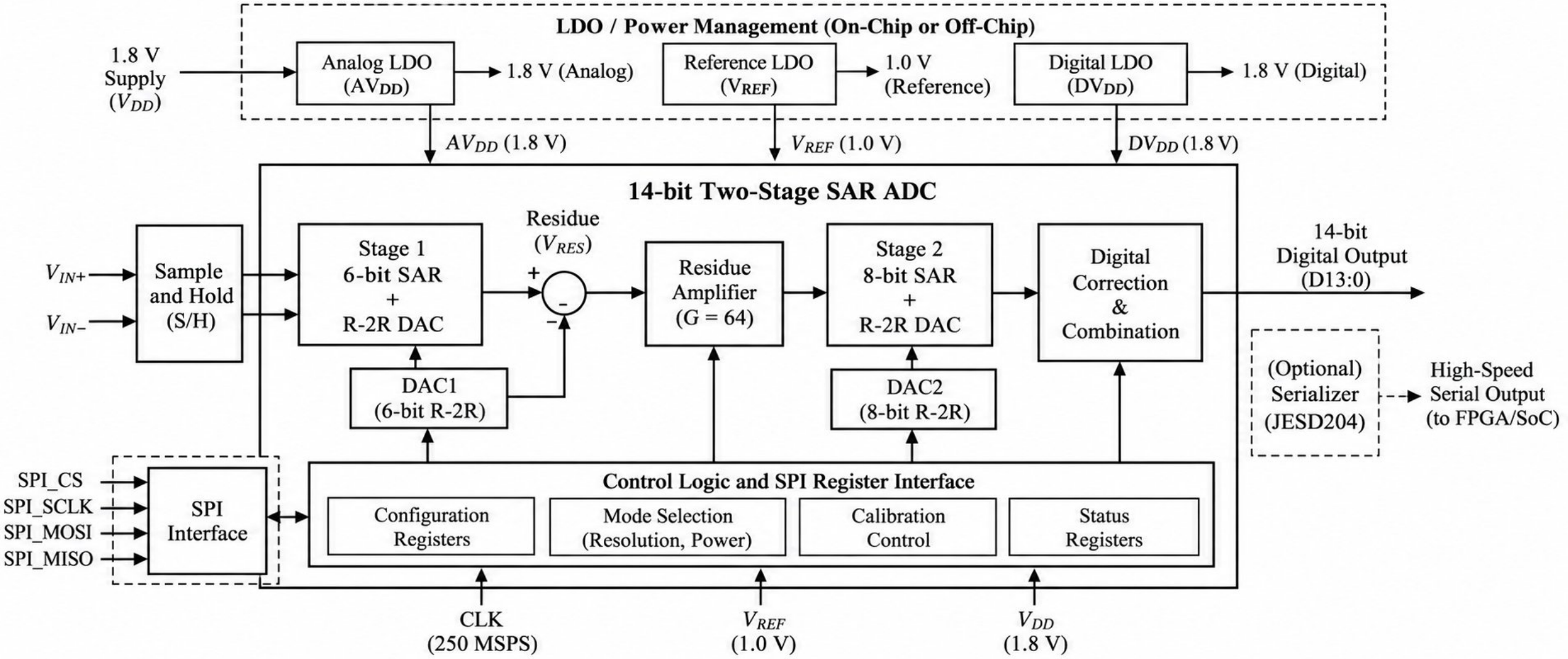
ADC Design Challenges



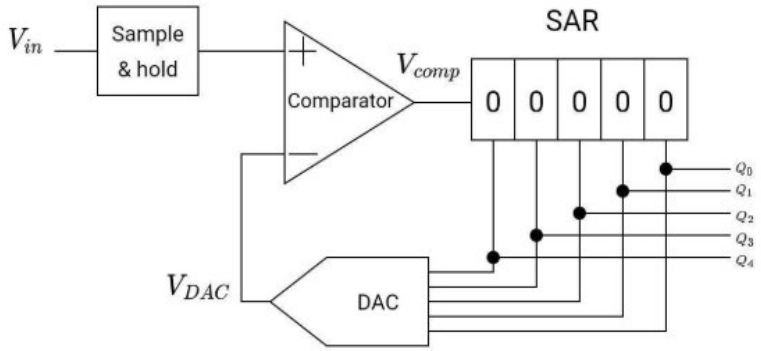
ADC Configuration with SPI

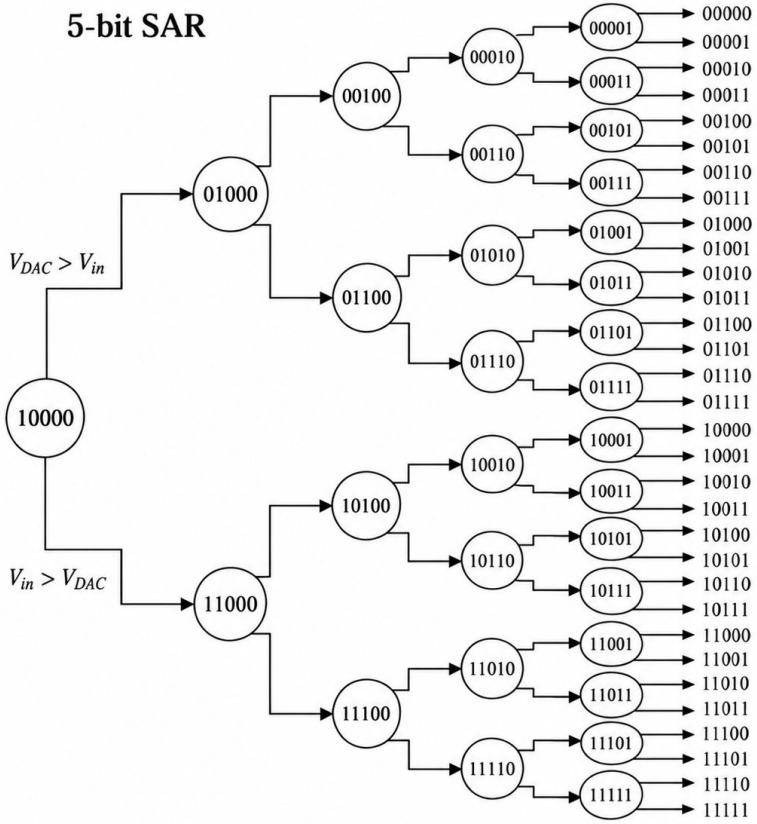


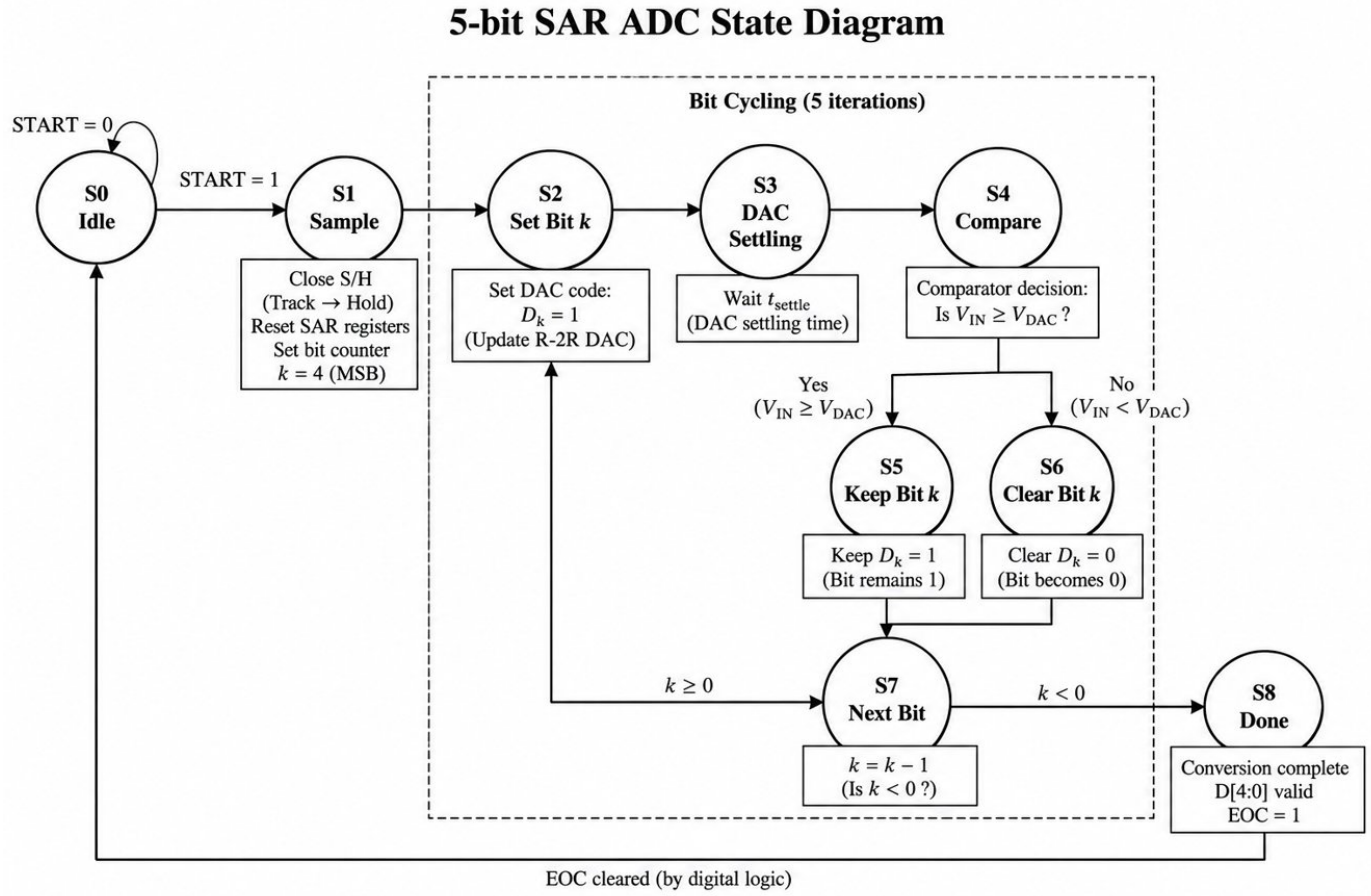
ADC IP


LDO / Power Management (On-Chip or Off-Chip)
1.8 V Supply (V_{DD})
Analog LDO (AV_{DD})
1.8 V (Analog)
Reference LDO (V_{REF})
1.0 V (Reference)
Digital LDO (DV_{DD})
1.8 V (Digital)
AV_{DD} (1.8 V)
V_{REF} (1.0 V)
DV_{DD} (1.8 V)
14-bit Two-Stage SAR ADC
V_{IN+}
V_{IN-}
Sample and Hold (S/H)
Stage 1 6-bit SAR + R-2R DAC
Residue (V_{RES})
Residue Amplifier (G = 64)
Stage 2 8-bit SAR + R-2R DAC
Digital Correction & Combination
14-bit Digital Output (D13:0)
DAC1 (6-bit R-2R)
DAC2 (8-bit R-2R)
(Optional) Serializer (JESD204)
High-Speed Serial Output (to FPGA/SoC)
SPI_CS
SPI_SCLK
SPI_MOSI
SPI_MISO
SPI Interface
Control Logic and SPI Register Interface
Configuration Registers
Mode Selection (Resolution, Power)
Calibration Control
Status Registers
CLK (250 MSPS)
V_{REF} (1.0 V)
V_{DD} (1.8 V)

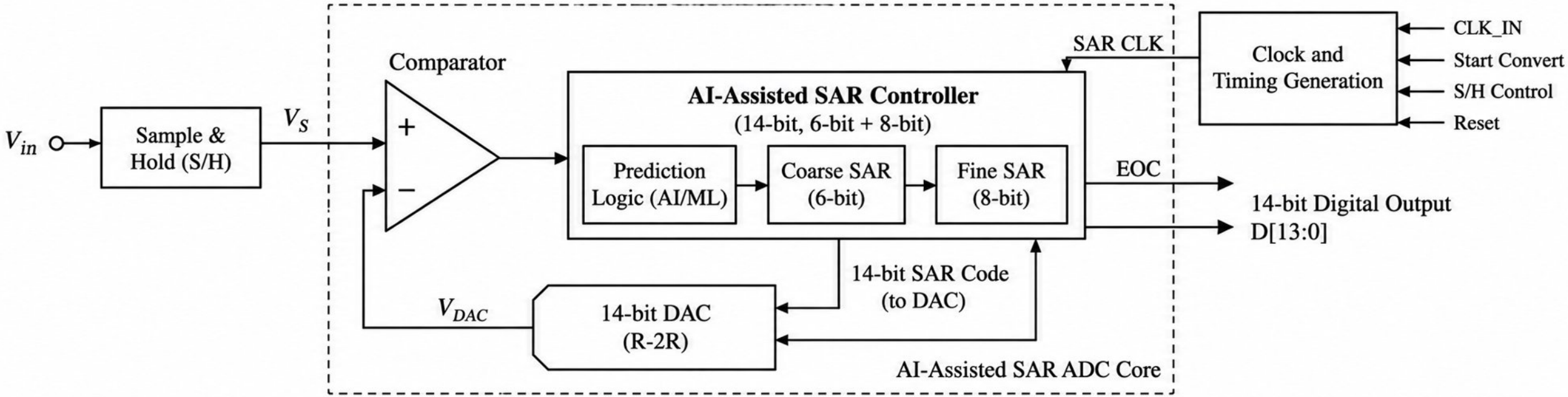
SAR Logic

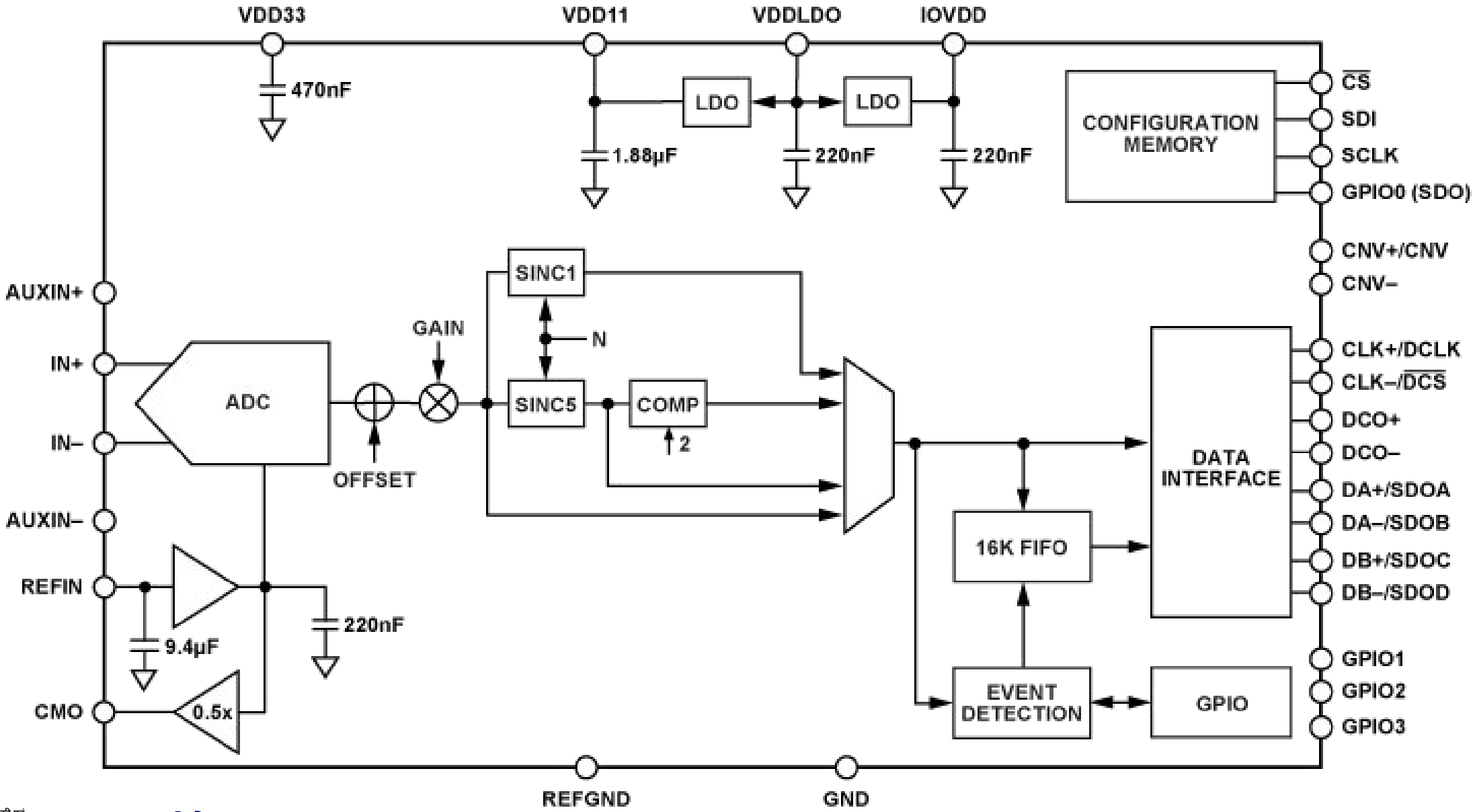

SAR
V_{in}
Sample & hold
Comparator
V_{comp}
0 0 0 0 0
Q_0
Q_1
Q_2
Q_3
Q_4
V_{DAC}
DAC


5-bit SAR
$V_{DAC} > V_{in}$
$V_{in} > V_{DAC}$
10000
01000
11000
00100
01100
10100
11100
00010
00110
01010
01110
10010
10110
11010
11110
00001
00011
00101
00111
01001
01011
01101
01111
10001
10011
10101
10111
11001
11011
11101
11111
00000
00001
00010
00011
00100
00101
00110
00111
01000
01001
01010
01011
01100
01101
01110
01111
10000
10001
10010
10011
10100
10101
10110
10111
11000
11001
11010
11011
11100
11101
11110
11111


5-bit SAR ADC State Diagram
Bit Cycling (5 iterations)
START = 0
START = 1
S0 Idle
S1 Sample
Close S/H (Track → Hold) Reset SAR registers Set bit counter k = 4 (MSB)
S2 Set Bit k
Set DAC code: D_k = 1 (Update R-2R DAC)
S3 DAC Settling
Wait t_{settle} (DAC settling time)
S4 Compare
Comparator decision: Is $V_{IN} \geq V_{DAC}$?
Yes ($V_{IN} \geq V_{DAC}$)
No ($V_{IN} < V_{DAC}$)
S5 Keep Bit k
Keep D_k = 1 (Bit remains 1)
S6 Clear Bit k
Clear D_k = 0 (Bit becomes 0)
$k \geq 0$
S7 Next Bit
$k = k - 1$ (Is $k < 0$?)
$k < 0$
S8 Done
Conversion complete D[4:0] valid EOC = 1
EOC cleared (by digital logic)

ADC IP


V_{in}
Sample & Hold (S/H)
V_S
Comparator
+
−
AI-Assisted SAR Controller
(14-bit, 6-bit + 8-bit)
Prediction Logic (AI/ML)
Coarse SAR (6-bit)
Fine SAR (8-bit)
SAR CLK
Clock and Timing Generation
CLK_IN
Start Convert
S/H Control
Reset
EOC
14-bit Digital Output
D[13:0]
14-bit SAR Code
(to DAC)
V_{DAC}
14-bit DAC
(R-2R)
AI-Assisted SAR ADC Core


VDD33
VDD11
VDDLDO
IOVDD
470nF
LDO
LDO
1.88µF
220nF
220nF
CONFIGURATION MEMORY
CS
SDI
SCLK
GPIO0 (SDO)
CNV+/CNV
CNV–
AUXIN+
IN+
IN–
AUXIN–
REFIN
CMO
GAIN
SINC1
N
ADC
OFFSET
SINC5
COMP
2
DATA INTERFACE
CLK+/DCLK
CLK–/DCS
DCO+
DCO–
DA+/SDOA
DA–/SDOB
DB+/SDOC
DB–/SDOD
16K FIFO
9.4µF
220nF
0.5x
EVENT DETECTION
GPIO
GPIO1
GPIO2
GPIO3
REFGND
GND

5 mm × 5 mm, 49-ball CSP_BGA package

TOP VIEW
(Not to Scale)

	1	2	3	4	5	6	7
A	VDD11	VDD11	VDD11	CNV+	CNV–	IOVDD	CLK–/ $\overline{\text{DCS}}$
B	REFGND	REFGND	VDDLDO	DCO–	DCO+	IOGND	CLK+/ DCLK
C	REFGND	REFGND	GND	GND	GND	DB+/ SDOC	DA+/ SDOA
D	IN+	AUXIN+	GND	GND	GND	DB–/ SDOD	DA–/ SDOB
E	IN–	AUXIN–	GND	GND	GND	GPIO1	SCLK
F	GPIO2	GPIO3	VDD33	REFIN	REFGND	REFGND	SDI
G	$\overline{\text{CS}}$	CMO	VDD33	REFIN	REFGND	DNC	GPIO0

GROUND PINS
DIGITAL I/O PINS
ANALOG INPUT PINS
POWER SUPPLY PINS
ANALOG OUTPUT PINS
DO NOT CONNECT

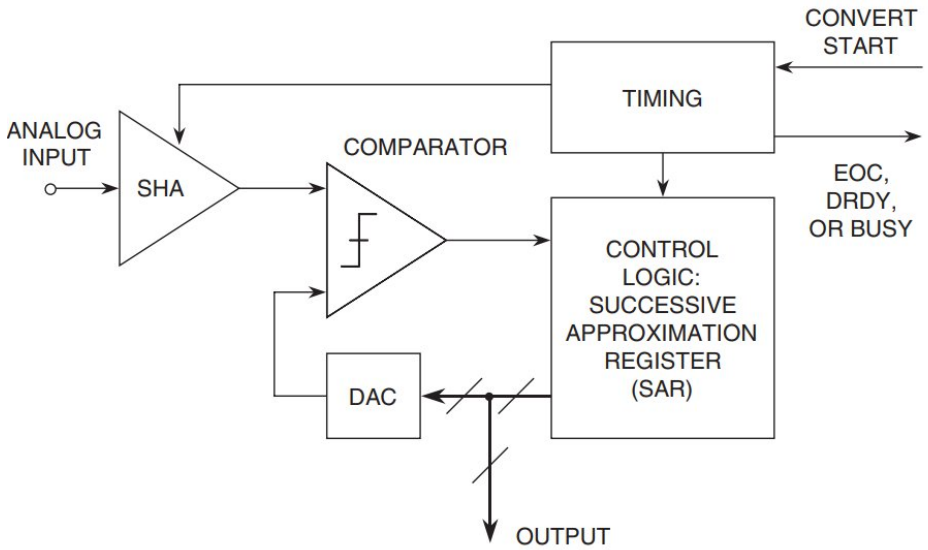
002

Pin group	Examples	Function
Analog input	IN+, IN–	Main differential ADC input
Auxiliary input	AUXIN+, AUXIN–	Auxiliary analog input
Reference	REFIN, REFGND	ADC reference
Conversion	CNV+, CNV–	Conversion control
Clock	CLK+, CLK–	ADC clock
Digital data	DA±, DB±	LVDS data interface
Data clock	DCO±	LVDS data clock
SPI	CS, SCLK, SDI	Configuration/control
Power	VDD11, VDD33, IOVDD	Supply rails
Ground	GND, RFEGND, IOGND	Ground/return paths
GPIO	GPIO0–GPIO3	Programmable digital fun.
Analog output	CMO	Common-mode output
DNC	DNC	Do not connect

- 14-bit resolution, no missing codes
- Throughput: 40MSPS, 48.21ns conversion latency
- Noise spectral density: 24.18nV/√Hz,158.9dBFS/Hz
- Low 1/f, low frequency noise (0.1Hz to 10Hz): 396nV rms
- Low Power:85mW typical at 40MSPS
- INL: ±12ppm (typ), ±16ppm (max)
- Dynamic range: 85.85dBFS
- SNR/THD
  - 85.34dB (typ)/−110.4dB (typ) at f_{IN} = 1kHz
  - 85.27dB (typ)/−101.3dB (typ) at f_{IN} = 1MHz
- Easy Drive
  - 6V p-p differential input range
  - Continuous signal acquisition
  - Linearized, 5µA/MSPS input current
- Integrated, low-drift reference buffer and decoupling
- Integrated V_{CM} generation
- Digital features and data interface
  - Conversion result FIFO, 16K sample depth
  - Digital averaging filter with up to 2^{10} decimation
- SPI configuration
- Configurable data interface
  - Single lane, DDR, serial LVDS, 560Mbps per lane
  - Dual lane, DDR, serial LVDS, 280Mbps per lane
  - Single/quad lane SPI data interface
- Package
  - 49-ball, 5mm × 5mm CSP_BGA, 0.65mm pitch
  - Integrated supply decoupling capacitors
- Operating temperature range: −40°C to +85°C

14-bit SAR ADC Design

What are we designing in this workshop ?



- Sample & Hold
- Comparator
- DAC
- SAR
- PWM
- OPAMP

Design Flow